

AmbientTransfer: Presence Enhancement by Converting Video Ambient to Users' Somatosensory Feedback

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ABSTRACT

Haptic feedback can improve users' presence when watching immersive videos. In this work, we present our "AmbientTransfer" framework for demonstrating how to "embed" users in the video ambient with somatosensory feedback. Particularly, AmbientTransfer can obtain and convert video ambient factors, e.g., rain intensities, into dynamic haptic feedback by slight electrostimulation in various levels. Then AmbientTransfer maps various levels of haptic feedback to different body parts of users wearing the emerging electrostimulation suit. Preliminary experimental results show that AmbientTransfer can considerably enhance the users' presence. We believe that AmbientTransfer is worth of further exploration for exploiting its full potentials.

Index Terms: Human-centered computing—Empirical studies in HCI; Human-centered computing—Virtual reality

1 INTRODUCTION

In recent years, the immersive video has been increasingly popular for its feature of offering presence to users. Relevant research shows that the users' presence can be enhanced by haptic feedback [1]. The emergence of somatosensory equipment offers a new dimension and new possibilities for haptic enhancement for users watching immersive videos. For example, the novel Teslasuit [2] is a smart textile suit that can provide full-body haptic and thermal feedback through electrostimulation. With Teslasuit, it has become feasible to simulate some "perceivable" ambient factors in the immersive video, e.g., the user's feel of being in the rainy, snowing, windy, warm conditions and so on [3]. If the above is achieved, it is expected that users can better immerse or even "embed" in the ambient of the video scene, i.e., achieving "ambient embeddedness" just like being the characters in the video scene.

However, there is still no framework that can obtain and "transfer" the video ambient to users wearing the somatosensory suit with haptic feedback, synchronized with the immersive video. Therefore, in this work, we present the AmbientTransfer framework (Figure 1) to fill this research gap. There are multiple ambient factors such as weather, temperature, object motion, etc., that could be obtained or estimated from videos. In this work, we demonstrate our idea and the feasibility of AmbientTransfer by focusing on the video scene with rainy ambient. To achieve our goal, we first obtain the rain intensities from the immersive video, generate slight electrostimulation in various levels according to the rain intensities, and then distribute various electrostimulation to different body

parts of users wearing Teslasuit (See Section 3 for details). By this means, we intend to "embed" the user in the ambient of the video scene for presence enhancement. To the best of our knowledge, our work is the first exploration in converting the video ambient, e.g., rain intensities in this research, into electrostimulation-based dynamic haptic feedback for users wearing somatosensory suit.

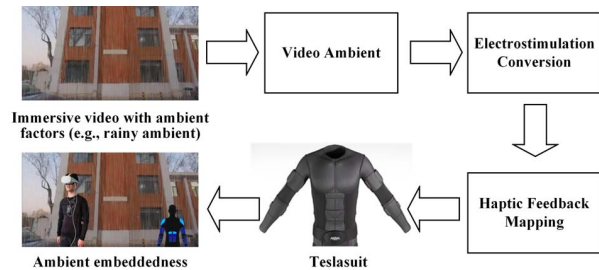


Figure 1. The framework of AmbientTransfer.

2 RELATED WORK

In related literature, Han et al. [4] proposed the "haptic around" tactile feedback approach, which can realize visual, auditory, and tactile perception in a virtual environment. Kishishita et al. [5] proposed a first-person VR game with a soft exoskeleton, which enables the interaction between the user and the object through an additional force feedback component. Yem et al. [6] proposed a method of inducing electrostimulation at the fingertips, which can present to the user the feeling of softness, hardness and stickiness of the touching objects. However, all the above researches (and similar work) are mainly designed to interact with virtual objects or provide haptic feedback without sufficiently considering the "ambient" in immersive video. At present, there is still no approach that can achieve "ambient embeddedness" by transferring the video ambient with novel haptic suits like Teslasuit.

3 METHODOLOGY

To obtain testing videos with rainy ambient for demonstration purpose, we generate 4 levels of rain intensities, including no rain ($\rho=0$), light rain ($\rho=5000$), moderate rain ($\rho=12000$), and heavy rain ($\rho=20000$), in our captured immersive video with the Adobe After Effects software. On the other hand, for natural videos that are without rain intensity parameters available, the work of Bossu [7] and Haurum [8] can be used to detect and estimate the rain intensity from the video. Subsequently, we can convert the rain intensities into various haptic levels. The Teslasuit can generate electrostimulation of up to 15mA, which can be programmed from 0 to 100 intensities. We set the maximum stimulation intensity to 10mA to ensure the participants' safety. Specifically, we calculate the electric current simulating the haptic feedback of rain as:

$$I(t) = kI_{\max} \rho(t) / \rho_{\max} \quad (1)$$

where $I(t)$ is the electric current at time t , k is the amplitude of the vibration, which we randomly choose from the range $[0.8, 1.2]$ to achieve realism, $I_{\max}$ is the maximum current, $\rho(t)$ is the rain intensity at time t , and $\rho_{\max}$ is the maximum rain intensity. In

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addition, to simulate the feeling of discontinuity when it rains, we consider that the frequency of the electrical stimulation is positively correlated with the rain intensity.

The Teslasuit's jacket [2] has 48 channels of electric currents, and each channel can be individually programmed. Among them, there are 8 channels on the back, 8 channels on the front, and 16 channels on each of the left and right arms. For demonstration purpose, we assume that the user is standing or sitting and therefore in this case, his/her shoulders become the direct contact area of the raindrops while other body parts are considered as indirect contact areas. To attempt simulating the natural feel of being in the rain, we design a simple-yet-effective scheme to unequally distribute the converted electrostimulation to the different parts of the user body in 8 intensity levels. For each indirect contact area (corresponding to a stimulation channel), the intensity level of the electrostimulation decreases as their distances to the direct contact areas increase. For a given intensity level, we assume that the magnitude of the haptic feedback decays exponentially as the distance to the direct contact areas increases. Then the magnitude of the haptic feedback of the n^{th} intensity level can be calculated as:

$$I(n) = I \cdot e^{-\left(\frac{n(\ln I_{\max} - \ln I_{\min})}{N} + \ln(I/I_{\max})\right)} \quad (2)$$

where I is the haptic magnitude, i.e., the electric current intensity as calculated in equation (1), and $n \leq N$, where N is the number of intensity levels ($N=8$). $I_{\max}$ is the highest haptic magnitude and $I_{\min}$ is the lowest haptic magnitude. Then the Teslasuit is programmed to reflect the haptic feedback on different body parts with various levels of electrostimulation.

4 PRELIMINARY EXPERIMENTAL RESULTS

We conducted a preliminary experiment on the current version of AmbientTransfer. Specifically, we recruited 15 participants (all male) from a population of university students. Their average age is 24.3 years old (from 22 to 30, $SD=2.06$). In the experiment, the testing immersive videos are captured by an Insta360 ONE X2 panoramic camera and processed by Adobe After Effects for rainy ambient generation. The testing video is presented to participants in three conditions including 1) no haptic feedback (No Haptic); 2) haptic feedback equally distributed to all the body parts, i.e., $I_e = 0.5I(t)$ for each body part (Equal Haptic); and 3) dynamic haptic feedback generated according to different rain intensities and unequally distributed to different body parts (AmbientTransfer). Each participant watches the captured immersive videos with the Oculus Quest 2 head-mounted display. The participants are then required to fill out the Igroup presence questionnaire (IPQ) [9] to report their presence experience. The IPQ results depicted in Figure 2 shows that AmbientTransfer yields stronger presence than the two baselines, particularly from the "experienced realism" perspective. Figure 3 shows some samples of the testing videos and the haptic feedback mapping to the user's different body parts.

5 CONCLUSION AND FUTURE WORK

We propose the novel AmbientTransfer framework to demonstrate the feasibility of transferring video ambient, which could be obtained or detected from video, to users wearing somatosensory suit with body-level haptic feedback. This preliminary version of AmbientTransfer is limited in only dealing with the rainy ambient. However, other ambient factors, such as snowing conditions, could be converted to electrostimulation in a similar way. Moreover, the current AmbientTransfer assumes the user's standing or sitting pose in a fixed position. In our future work, we plan to develop a more comprehensive version of AmbientTransfer that can adapt to different poses and positions of the users.

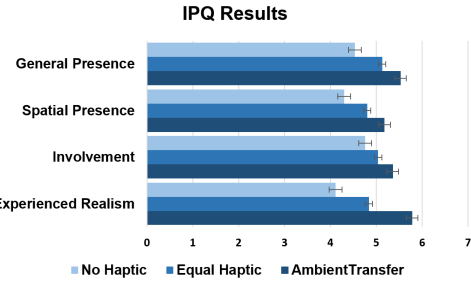


Figure 2: Igroup Presence Questionnaire (IPQ) results.



Figure 3: Screenshots of the testing videos and sample illustrations of the haptic mapping.

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